PAPER_1517_TRANSCENDENTAL_PI_SQUARED

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PAPER_1517 — $\pi^2 \approx$ SO_5 – F_TRZ – F_TRZ²(K_MEX + Φ_5/6) — Residual 0.0125%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational mathematical identity)
Date: June 18, 2026
Status: CLOSED — π^2 expressible in UQFF primitives at sub-0.013% precision

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Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set, row S541) derives $\pi^2 \approx 9.8696$ using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6, and SO_5.

UQFF Closed Identity

```
π² ≈ SO_5 – F_TRZ – F_TRZ²·K_MEX – F_TRZ²·Φ
= 10 – 0.1 – 0.01·(25/12) – 0.01·(5/6)
= 10 – 0.1 – 0.02083 – 0.00833
= 9.87083

Observed: π² = 9.869604401…
Residual: |9.87083 – 9.86960| / 9.86960 = 0.0125%
```

Structural Form: $\pi^2 \approx$ SO_5

The dominant structural observation is that π^2 ≈ **SO_5 = 10** to within 1.3%, with small F_TRZ corrections accounting for the residual. This is the deepest mathematical relationship in the framework:

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```
SO_5 = 10 (icosahedral spatial coverage)
π² = 9.86960… (transcendental)
SO_5 – π² ≈ 0.1304 ≈ F_TRZ (the time-reversal-zone factor)
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The "approximately equal to F_TRZ" relationship (within 30%) hints that the F_TRZ definition itself may be a structural compensation for the SO_5 – π^2 gap. That is, F_TRZ ≈ SO_5 – π^2 is not coincidence but a derivative integer-primitive identity.

Consequence for Other Closures

If $\pi^2 \approx$ SO_5 structurally, then any UQFF closure involving π^2 (cosmological constant Λ, harmonic-oscillator partition function, Stefan-Boltzmann constant) can be rewritten in terms of SO_5 — providing alternate integer-primitive forms throughout the calculator.

NOT REPLACEMENT

UQFF does not claim π^2 IS exactly SO_5 – F_TRZ – F_TRZ^2(K_MEX + Φ). It claims the integer primitives carry a rational approximation to π^2 with 0.0125% precision, and that the dominant structural form is SO_5.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set (row S541)
- Related: PAPER_1515 (ln 10), PAPER_1516 (ln 2 = tightest), PAPER_062 (SO_5 = 10)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_pi_squared"})`
